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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/195,589	03/08/2021	Yu Fang	SONA-0010-U01	2972

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EXAMINER	
LIN, JSING FORNG	

ART UNIT	PAPER NUMBER
2446	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 17/195,589	Applicant(s) Fang et al.	
	Examiner JSING FORNG LIN	Art Unit 2446	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 3/8/2021.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) 10-20 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-9 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☑ Claim(s) 1-20 are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☑ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

This Action is in response to application filed on 3/8/2021.

Claims 1-20 are pending in this application.

Claims 10-20 are withdrawn.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on 3/8/2021, 2/7/2022 are in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Claim Objections

Claims 5-9 are objected to because of the following informalities:

- Claims “5, 6, 7, 8, 9” should be “4, 5, 6, 7, 8”.

Appropriate correction is required.

Election/Restrictions

Applicant’s election without traverse of group I, claims 1-9 in the telephone communication on 8/24/2021.

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the claims at issue are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., In re Berg, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); In re Goodman, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); In re Longi, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); In re Van Ornum, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); In re Vogel, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and In re Thorington, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on a nonstatutory double patenting ground provided the reference application or patent either is shown to be commonly owned with this application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO internet Web site contains terminal disclaimer forms which may be used. Please visit http://www.uspto.gov/forms/. The filing date of the application will determine what form should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all

requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to http://www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

Claims 1-9 are rejected on the ground of nonstatutory double patenting as being unpatentable over claim 1 of U.S. Application No. 17,027,187. Although the claims at issue are not identical, they are not patentably distinct from each other because claims in the patented application anticipated claims in the present application.

U.S. Application No. 17,027,187	Present Application No. 17,195,589
Claim 1. A system, comprising: a vehicle having at least one network zone; a policy manager circuit structured to interpret a policy comprising a local domain name server (DNS), an authorization description, and a firewall configuration description; a configuration circuit structured to configure a gatekeeper interface circuit in response to the policy; and a gatekeeper	Claim 1. An apparatus comprising: a remote access execution circuit structured to interpret a remote access request value from a requesting device, the remote access request value comprising at least one of a requested vehicle property or a vehicle function value; a property translation circuit structured to determine at least one of: a property request value in response to the at least

interface circuit interposed between the at least one network zone and an external communication portal selectively couplable to an external device, and	one requested vehicle property; or an actuator command value in response to the vehicle function value; a parameter acquisition circuit structured to interpret a plurality of vehicle parameter values in response to the property request value; a remote operation circuit structured to provide the actuator command value to an endpoint of a network zone of a vehicle;
further structured to regulate communications between end points of the at least one network zone and the external communication portal.	a parameter conditioning circuit structured to generate, in response to the property request value, vehicle property data from the plurality of vehicle parameter values, the vehicle property data corresponding to at least one the requested vehicle property; and wherein the remote access execution circuit is further structured to transmit the vehicle property data to the requesting device.

As for the dependent claims 2-9 of the instant application, the scope is covered by claim 1 of the patented application in view of Lim (US 2016/0133063 A1) and Barrett (US 2019/0394089 A1) which describe exchange message between intra-domain.

CLAIM INTERPRETATION

The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

This application includes one or more claim limitations that use the word "means" or "step" or "structured" but are nonetheless not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph because the claim limitation(s) recite(s) sufficient structure, materials, or acts to entirely perform the recited function. Such claim limitation(s) is/are:

- ... execution circuit structured to interpret ... claim 1 limitation 1
- ... translation circuit structured to determine ... claim 1 limitation 2
- ... acquisition circuit structured to interpret ... claim 1 limitation 3
- ... operation circuit structured to provide ... claim 1 limitation 4
- ... conditioning circuit structured to generate ... claim 1 limitation 5

Examiner interpret circuit as hardware with specific structure corresponding to specification (for example paragraph 0450, 0841, 0857)

Because this/these claim limitation(s) is/are **not** being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it/they is/are **not** being interpreted to cover only the corresponding structure, material, or acts described in the specification as performing the claimed function, and equivalents thereof.

If applicant intends to have this/these limitation(s) interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to remove the structure, materials, or acts that performs the claimed function; or (2) present a sufficient showing that the claim limitation(s) does/do not recite sufficient structure, materials, or acts to perform the claimed function.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly

owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claims 1-3, 5-9 are rejected under 35 U.S.C. 103 as being unpatentable over Lim et al (US 2016/0133063 A1) in view of Barrett et al (US 2019/0394089 A1).

Regarding claim 1, Lim teaches an apparatus comprising: a remote access execution circuit structured to interpret a remote access request value from a requesting device, the remote access request value comprising at least one of a requested vehicle property or a vehicle function value (Lim fig. 6 item #S610, #620 receive diagnosis request message (VID: vehicle ID, N: number of actuator), VID agreement?, [0069]);

a property translation circuit structured to determine at least one of: a property request value in response to the at least one requested vehicle property; or an actuator command value in response to the vehicle function value (Lim [0070] if VID match, perform diagnostic actuator logic to perform the number N of actuators to be diagnosed to sequentially);

a parameter acquisition circuit structured to interpret a plurality of vehicle parameter values in response to the property request value; a remote operation circuit structured to provide the actuator command value to an endpoint of a vehicle (Lim [0073-0075] diagnose actuator based on the receive reference power and tolerance range; drives the actuator to be diagnosed and starts a predetermined timer (S633) to start to measure the power consumption up to a predetermined time (S634));

a parameter conditioning circuit structured to generate, in response to the property request value, vehicle property data from the plurality of vehicle parameter values, the vehicle property data corresponding to at least one the requested vehicle property; and wherein the remote access execution circuit is further structured to transmit the vehicle property data to the requesting device (Lim [0076-0077] notify the diagnostic terminal of actuator diagnostic result, such as whether to instruct the driving of the actuator, the power change measured by the measurement unit, the comparison result of the power change with the reference power value stored in the storage, and the change trend of the measurement current).

However, Lim does not explicitly teach to provide the command value to an endpoint of a network zone of a vehicle;

Barrett teaches to provide the command value to an endpoint of a network zone of a vehicle (Barrett [0037][0042] intra-domain firewalling vehicle communication, firewall safeguard traffic passing in between domain based on rules; domain end points may include actuator ECUs)

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify Lim in view of Barrett to provide the command value to an endpoint of a network zone of a vehicle.

One of ordinary skill in the art before the effective filing date of the claimed invention would have been motivated because it would have enable the system to perform message processing more efficiently, i.e. to enable communication based on rules (Barrett [0042]).

Regarding claim 2, Lim-Barrett teaches the apparatus of claim 1, further comprising a converged network device (CND) structured to regulate communications between a first network zone having a first network endpoint and a second network zone having a second network endpoint, wherein at least a portion of the plurality of vehicle parameter values are generated by each of the first network endpoint and the second network endpoint (Barrett [0034][0037][0042] intra-domain firewalling vehicle network to provide communication in between network based on rules).

Regarding claim 3, Lim-Barrett teaches the apparatus of claim 1, further comprising: a converged network device (CND) structured to regulate communications between a first network zone having a first network endpoint and a second network zone having a second network endpoint and comprising the network zone of the vehicle; wherein the first network endpoint provides at least a portion of the plurality of vehicle parameter values; and wherein the second network endpoint comprises an actuator responsive to the actuator command value (Barrett [0034][0042] network zone end point can be actuator; Lim [0072]).

Regarding claim 5, Lim-Barrett teaches the apparatus of claim 1, wherein the property translation circuit is further structured to determine the actuator command value by performing at least one operation selected from the operations consisting of: determining the actuator command value as a sequence of actuator commands corresponding to a diagnostic test operation; determining the actuator command value as a sequence of actuator commands corresponding to a remote control operation; or

determining the actuator command value as at least one actuator command responsive to the vehicle function value (Lim [0070][0073-0075] diagnose actuator based on the receive reference power and tolerance range and perform diagnostic actuator logic to perform the number N of actuators to be diagnosed to sequentially).

Regarding claim 6, Lim-Barrett teaches the apparatus of claim 1, wherein the property request value includes at least one of: a vehicle speed value; a prime mover speed value; a prime mover torque value; a user actuated vehicle feature value; or a vehicle location value (Lim [0077] based on received power reference and notify results power change with reference power value).

Regarding claim 7, Lim-Barrett teaches the apparatus of claim 1, wherein the property request value includes at least one of: a network utilization value for a network zone of the vehicle; a raw network message from a network zone of the vehicle; a network address for an endpoint on a network zone of the vehicle; a memory storage description of a controller of the vehicle; a value from an end point on a controller area network (CAN); a value from an end point on a local interconnect network (LIN); or an intermediate control value (Barrett [0037] intra-network domain includes LIN, CAN or FlexRay)

Regarding claim 8, Lim-Barrett teaches the apparatus of claim 1, further comprising: an automated operation circuit structured to interpret an automated operation value comprising an automated operation description for the vehicle (Lim

[0073-0075]); an automation manager circuit structured to determine a trigger description value in response to the automated operation value, the trigger description value comprising a trigger condition value and a trigger response value (Barrett [0088][0095] trigger based on analysis); a trigger evaluation circuit structured to determine a trigger event occurrence in response to the trigger condition value and at least one vehicle data value; and a trigger execution circuit structured to execute a trigger response in response to the trigger event occurrence (Barrett [0088][0095] trigger to perform action based on received description).

Regarding claim 9, Lim-Barrett teaches the apparatus of claim 1, wherein the property request value includes at least one of: a fault condition value; a fault count value; a diagnostic parameter value; a fault confirmation value; a diagnostic confirmation value; a fault intermediate value; or a diagnostic intermediate value (Lim [0012][0015] diagnostic request/result includes reference power value, timer expire, and out of tolerance range; [0069] notify negative result if VID not match) .

Additional References

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

a. Lee et al., US 9,355,507 B1: System and method for collecting data of vehicle.
b. King et al., US 10,650,621 B1: Interface with a vehicular controller area network.

c. Penilla et al., US 20150210287 A1: VEHICLES AND VEHICLE SYSTEMS FOR PROVIDING ACCESS TO VEHICLE CONTROLS, FUNCTIONS, ENVIRONMENT AND APPLICATIONS TO GUESTS/PASSENGERS VIA MOBILE DEVICES.

d. SCHOCH, USS 2019/0356574 A1: Motor vehicle comprising an internal data network and method for operating the motor vehicle.

e. Kim et al., US 20160255154 A1: VEHICLE SECURITY NETWORK DEVICE AND DESIGN METHOD THEREFOR.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEVE LIN whose telephone number is 571-272-5137. The examiner can normally be reached on Monday – Friday 7:30 AM – 5:00 PM.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Brian Gillis can be reached on 571-272-7952. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a

USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/S. L./
Examiner, Art Unit 2446

/MICHAEL A KELLER/
Primary Patent Examiner, Art Unit 2446